

MCQS

CS614 midterm preparation file

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Gracious, Most merciful



CS614

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1. What is the primary purpose of a data warehouse?

- A) Online Transaction Processing (OLTP)
- B) Decision Support
- C) Real-time data capture
- D) Network Management

Answer: B) Decision Support

2. Which of the following best describes ETL?

- A) Extract, Transform, Load
- B) Evaluate, Transfer, Load
- C) Extract, Transport, Log
- D) Evaluate, Transform, Log

Answer: A) Extract, Transform, Load

3. In a star schema, what connects fact tables to dimension tables?

- A) Surrogate keys
- B) Foreign keys
- C) Primary keys
- D) Composite keys

Answer: B) Foreign keys

4. What type of schema consists of a central fact table surrounded by normalized dimension tables?

- A) Star Schema
- B) Snowflake Schema
- C) Galaxy Schema
- D) Relational Schema

Answer: B) Snowflake Schema

5. Which one is NOT a characteristic of a data warehouse?

- A) Subject-oriented
- B) Non-volatile
- C) Real-time updated
- D) Time-variant

Answer: C) Real-time updated

6. A data mart is typically designed for which of the following?

- A) A single business process
- B) Enterprise-wide analysis
- C) Network management
- D) Data cleansing

Answer: A) A single business process

7. Which of the following is a major benefit of dimensional modeling?

- A) Complex joins
- B) Easier understanding by end-users
- C) Normalized design
- D) Minimal data redundancy

Answer: B) Easier understanding by end-users

8. What is a surrogate key in the context of a data warehouse?

- A) A natural primary key
- B) A generated, unique identifier
- C) A composite foreign key
- D) A shared key between tables

Answer: B) A generated, unique identifier

9. Which of the following is used for aggregating data in a data warehouse?

- A) OLTP
- B) OLAP
- C) ETL
- D) SQL

Answer: B) OLAP

10. What does the term 'slice and dice' refer to in OLAP?

- A) ETL process
- B) Data transformation
- C) Multidimensional data analysis
- D) Query optimization

Answer: C) Multidimensional data analysis

11. Which type of OLAP stores data in relational databases?

- A) ROLAP
- B) MOLAP
- C) HOLAP
- D) DOLAP

Answer: A) ROLAP

12. The process of improving data quality in ETL is called:

- A) Data extraction
- B) Data profiling
- C) Data cleansing
- D) Data loading

Answer: C) Data cleansing

13. Which schema design minimizes redundancy?

- A) Star Schema
- B) Snowflake Schema
- C) Fact Constellation
- D) Flat Schema

Answer: B) Snowflake Schema

14. Fact tables usually contain:

- A) Descriptive data
- B) Aggregate data
- C) Key performance indicators (KPIs)
- D) Transactional data

Answer: D) Transactional data

15. Which tool is commonly used for ETL processes?

- A) Tableau
- B) Informatica
- C) MongoDB
- D) Power BI

Answer: B) Informatica

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16. Which is an example of a time-variant characteristic in data warehousing?

- A) Employee ID
- B) Customer Name
- C) Sales data over years
- D) Email address

Answer: C) Sales data over years

17. In a data warehouse, metadata refers to:

- A) Raw data
- B) Summary data
- C) Data about data
- D) Aggregated data

Answer: C) Data about data

18. What type of relationship exists between a fact table and dimension tables?

- A) Many-to-many
- B) One-to-many
- C) One-to-one
- D) Many-to-one

Answer: B) One-to-many

19. Which technique is used to reduce the size of a data warehouse?

- A) Data cleansing
- B) Data summarization
- C) Data normalization
- D) Data fragmentation

Answer: B) Data summarization

20. Which of the following is an example of a dimensional attribute?

- A) Product ID
- B) Product Name
- C) Transaction Amount
- D) Invoice Number

Answer: B) Product Name

21. Which type of schema uses multiple fact tables?

- A) Star Schema
- B) Snowflake Schema
- C) Fact Constellation Schema
- D) Flat Schema

Answer: C) Fact Constellation Schema

22. What is a conformed dimension?

- A) Dimension shared across fact tables
- B) A normalized dimension
- C) A temporary table
- D) A transactional dimension

Answer: A) Dimension shared across fact tables

23. What is granularity in a fact table?

- A) Level of detail stored
- B) Number of attributes
- C) Type of keys
- D) Storage format

Answer: A) Level of detail stored

24. Which of the following is a type of OLAP operations?

- A) Aggregation
- B) Drill-down
- C) Data cleansing
- D) Transaction processing

Answer: B) Drill-down

25. In dimensional modeling, a degenerate dimension contains:

- A) No attributes
- B) Only keys
- C) Summary data
- D) Multiple hierarchies

Answer: A) No attributes

26. Which architecture layer is unique to a data warehouse?

- A) Presentation layer
- B) Data staging layer
- C) Application layer
- D) Network layer

Answer: B) Data staging layer

27. What is the purpose of an OLAP cube?

- A) Store raw data
- B) Optimize transaction processing
- C) Facilitate multidimensional analysis
- D) Normalize relational data

Answer: C) Facilitate multidimensional analysis

28. Which component ensures data consistency in a data warehouse?

- A) ETL
- B) Metadata
- C) OLAP
- D) Data Mart

Answer: A) ETL

29. Which term refers to splitting large datasets into smaller parts?

- A) Partitioning
- B) Sharding
- C) Clustering
- D) Indexing

Answer: A) Partitioning

30. A slowly changing dimension is one that:

- A) Frequently changes
- B) Rarely changes
- C) Never changes
- D) Changes unpredictably

Answer: B) Rarely changes

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31. In a star schema, the central table is called the:

- A) Fact table
- B) Dimension table
- C) Lookup table
- D) Aggregation table

Answer: A) Fact table

32. Which of the following is NOT a type of slowly changing dimension (SCD)?

- A) Type 1
- B) Type 2
- C) Type 3
- D) Type 4

Answer: D) Type 4

33. What is the process of organizing data into layers called?

- A) ETL
- B) Data Normalization
- C) Data Staging
- D) Data Aggregation

Answer: C) Data Staging

34. Which type of join is used in ETL processes to match source and target data?

- A) Inner Join
- B) Outer Join
- C) Left Join
- D) Full Join

Answer: A) Inner Join

35. In a data warehouse, which term describes data at the lowest level of granularity?

- A) Summary data
- B) Atomic data
- C) Aggregated data
- D) Derived data

Answer: B) Atomic data

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36. Which of the following is true about OLAP?

- A) It supports transaction processing
- B) It focuses on historical data
- C) It is used for real-time operations
- D) It supports write-heavy workloads

Answer: B) It focuses on historical data

37. What is the main purpose of metadata in a data warehouse?

- A) Store raw data
- B) Document data structures
- C) Improve query speed
- D) Provide real-time analytics

Answer: B) Document data structures

38. Which of the following best describes MOLAP?

- A) Uses relational databases
- B) Uses multidimensional data storage
- C) Combines ROLAP and MOLAP
- D) Stores unstructured data

Answer: B) Uses multidimensional data storage

39. What is a conformed fact?

- A) A fact shared across multiple schemas
- B) A normalized fact table
- C) A temporary summary
- D) A fact with no dimensions

Answer: A) A fact shared across multiple schemas

40. Which type of data model is primarily used in data warehousing?

- A) Hierarchical
- B) Network
- C) Dimensional
- D) Object-oriented

Answer: C) Dimensional

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41. What is the term for data duplication in a data warehouse?

- A) Redundancy
- B) Fragmentation
- C) Indexing
- D) Aggregation

Answer: A) Redundancy

42. Which tool is commonly used for OLAP operations?

- A) Power BI
- B) MongoDB
- C) Oracle SQL Developer
- D) Hadoop

Answer: A) Power BI

43. In a snowflake schema, dimensions are:

- A) Denormalized
- B) Normalized
- C) Flattened
- D) Aggregate-only

Answer: B) Normalized

44. What is the purpose of drill-through in OLAP?

- A) View summary data
- B) Access detailed data
- C) Aggregate data
- D) Clean data

Answer: B) Access detailed data

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45. Which type of dimension includes hierarchical data?

- A) Junk dimension
- B) Degenerate dimension
- C) Role-playing dimension
- D) Hierarchical dimension

Answer: D) Hierarchical dimension

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46. What is a junk dimension?

- A) Stores frequently changing data
- B) Combines unrelated attributes
- C) Contains derived metrics
- D) Stores metadata

Answer: B) Combines unrelated attributes

47. Which statement best describes ETL performance tuning?

- A) Optimizing data query execution
- B) Improving the speed of ETL jobs
- C) Compressing data in OLAP cubes
- D) Enhancing metadata readability

Answer: B) Improving the speed of ETL jobs

48. In data warehousing, a derived fact is:

- A) Calculated from other facts
- B) Directly extracted from source systems
- C) A normalized value
- D) A temporary metric

Answer: A) Calculated from other facts

49. Which OLAP operation changes the view level of data from detailed to summary?

- A) Drill-down
- B) Roll-up
- C) Pivoting
- D) Slicing

Answer: B) Roll-up

50. What is a primary advantage of using surrogate keys?

- A) Readability
- B) Performance optimization
- C) Unique identification across systems
- D) Consistency with source keys

Answer: C) Unique identification across systems

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51. Which of the following is an example of a role-playing dimension?

- A) Product Dimension
- B) Date Dimension used for Order Date and Delivery Date
- C) Customer Dimension
- D) Location Dimension

Answer: B) Date Dimension used for Order Date and Delivery Date

52. What is the key purpose of a data lake?

- A) Store only structured data
- B) Store raw and unstructured data
- C) Replace data warehouses
- D) Focus on real-time analytics

Answer: B) Store raw and unstructured data

53. Which of the following is true about a snowflake schema?

- A) More complex queries than star schema
- B) No redundancy in dimension tables
- C) Requires less storage space
- D) All of the above

Answer: D) All of the above

54. Which is an example of an additive measure in a fact table?

- A) Average sales
- B) Total revenue
- C) Customer count
- D) Conversion rate

Answer: B) Total revenue

55. Which of the following is a key feature of ROLAP?

- A) Precomputed summaries
- B) Dynamic aggregation using SQL queries
- C) Cube-based storage
- D) Real-time analysis

Answer: B) Dynamic aggregation using SQL queries

56. What is a shared dimension?

- A) A dimension unique to a single fact table
- B) A dimension shared across multiple schemas
- C) A dimension with duplicate attributes
- D) A temporary staging table

Answer: B) A dimension shared across multiple schemas

57. What does OLAP stand for?

- A) Online Analytical Processing
- B) Online Aggregation Processing
- C) Offline Analytical Processing
- D) Offline Aggregation Processing

Answer: A) Online Analytical Processing

58. Which database architecture is best for OLTP?

- A) Dimensional
- B) Normalized
- C) Flat file
- D) Hierarchical

Answer: B) Normalized

59. Which term describes the process of rearranging dimensions in an OLAP cube?

- A) Slicing
- B) Pivoting
- C) Roll-up
- D) Drill-down

Answer: B) Pivoting

60. Which type of dimension handles changing customer addresses?

- A) Slowly Changing Dimension
- B) Rapidly Changing Dimension
- C) Degenerate Dimension
- D) Junk Dimension

Answer: A) Slowly Changing Dimension

61. Which type of Slowly Changing Dimension (SCD) keeps the entire history of changes?

- A) Type 1
- B) Type 2
- C) Type 3
- D) Type 4

Answer: B) Type 2

62. Which of the following is an example of a semi-additive measure?

- A) Total Sales
 - B) Average Salary
 - C) Current Inventory Level
 - D) Number of Transactions
- Answer: C) Current Inventory Level

63. The granularity of a fact table refers to:

- A) The size of the database
 - B) The level of detail stored
 - C) The number of dimensions
 - D) The complexity of queries
- Answer: B) The level of detail stored

64. Which statement best describes an OLAP cube?

- A) It is a physical data structure in the database
 - B) It is a logical representation for multidimensional data
 - C) It is used for real-time updates
 - D) It is a flat structure used for queries
- Answer: B) It is a logical representation for multidimensional data

65. Which type of schema is most common in a data warehouse?

- A) Star Schema
 - B) Snowflake Schema
 - C) Fact Constellation Schema
 - D) Network Schema
- Answer: A) Star Schema

66. What is a degenerate dimension in a data warehouse?

- A) A dimension table with only one attribute**
- B) A dimension stored in a fact table**
- C) A dimension with hierarchical data**
- D) A dimension with multiple keys**

Answer: B) A dimension stored in a fact table

67. Which of the following is an operation commonly performed in ETL?

- A) Data Compression**
- B) Data Transformation**
- C) Data Indexing**
- D) Data Clustering**

Answer: B) Data Transformation

68. Which statement about a fact table is true?

- A) It contains descriptive information**
- B) It contains only numerical data**
- C) It connects dimension tables**
- D) It contains metadata**

Answer: C) It connects dimension tables

69. What does "non-volatile" mean in the context of a data warehouse?

- A) Data is frequently updated**
- B) Data is updated in real time**
- C) Data does not change once entered**
- D) Data is stored redundantly**

Answer: C) Data does not change once entered

70. Which of the following is an example of a measure in a fact table?

- A) Product Name**
- B) Sales Amount**
- C) Customer ID**
- D) Transaction Date**

Answer: B) Sales Amount

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71. What is the purpose of data aggregation in a data warehouse?

- A) Improve query performance
- B) Normalize data
- C) Remove duplicates
- D) Increase data volume

Answer: A) Improve query performance

72. Which type of schema uses denormalized dimension tables?

- A) Snowflake Schema
- B) Star Schema
- C) Flat Schema
- D) Fact Constellation Schema

Answer: B) Star Schema

73. Which of the following tools is NOT typically used for ETL?

- A) Informatica
- B) Talend
- C) Tableau
- D) Pentaho

Answer: C) Tableau

74. What is the purpose of a role-playing dimension?

- A) Store hierarchical data
- B) Serve multiple purposes in a schema
- C) Store unrelated attributes
- D) Combine measures from multiple fact tables

Answer: B) Serve multiple purposes in a schema

75. What is the main characteristic of a Type 1 Slowly Changing Dimension?

- A) Tracks historical data
- B) Updates data without keeping history
- C) Adds new rows for changes
- D) Uses a separate table for history

Answer: B) Updates data without keeping history

MIDTERM Important topic from 1 to 22 lectures

Core Topics to Focus On

1. Distributed Databases

Distributed database architectures

Data fragmentation, replication, and allocation

Query processing and optimization in distributed environments

2. Concurrency Control

Techniques: Two-Phase Locking (2PL), Time-Stamp Ordering

Deadlock detection and prevention

Distributed transaction management

3. Recovery Techniques

Distributed recovery protocols

Checkpointing and logging

Handling distributed failures

4. Data Replication

Consistency models

Synchronous vs. asynchronous replication

Conflict resolution strategies

5. Big Data and NoSQL Systems

Key-value stores, column-family stores, document stores

CAP theorem and its implications

Hadoop and MapReduce fundamentals (if part of the syllabus)

6. Performance Tuning and Optimization

Query optimization techniques

Indexing strategies

Cost-based optimization

7. Distributed Query Processing

Query decomposition and data localization

Semi-join and join optimization

8. Emerging Topics (if covered)

Cloud databases

Blockchain for distributed databases

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Additional Key Topics:

1. Distributed Hashing and Load Balancing

Consistent hashing and its application in distributed systems

Load balancing techniques and their significance in large-scale distributed systems

2. Transactions in Distributed Systems

ACID properties in a distributed context

Atomic commit protocols like 2PC (Two-Phase Commit) and Paxos consensus protocol

3. MapReduce and Parallel Processing

MapReduce programming model and its use in distributed data processing

Optimizing MapReduce jobs for large datasets

4. Replication and Fault Tolerance

Master-slave replication vs. peer-to-peer replication

Strategies to ensure fault tolerance and data durability in distributed systems

5. Quorum-Based Approaches

Majority quorum, read/write quorum in distributed systems

How quorum-based approaches handle consistency and availability

6. Sharding and Data Partitioning

Horizontal and vertical partitioning techniques

Sharding strategies for large databases and their challenges (e.g., re-sharding)

7. Eventual Consistency and CAP Theorem

Understanding eventual consistency and its trade-offs in distributed databases

CAP theorem and the implications of Consistency, Availability, and Partition tolerance

8. Data Warehousing and OLAP

Design and structure of data warehouses

OLAP (Online Analytical Processing) and its use in multi-dimensional analysis

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**"Marks ka tension? Bas
apne dil ko samjhao: 'Main
genius hoon!'"**

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1. What is a data warehouse?

- A) A collection of operational data
- B) A subject-oriented, integrated, time-variant, and non-volatile collection of data
- C) A system for OLTP transactions
- D) None of the above

Answer: B

2. Which schema consists of a single fact table with multiple dimension tables?

- A) Star schema
- B) Snowflake schema
- C) Fact constellation
- D) None of the above

Answer: A

3. ETL stands for:

- A) Extract, Transform, Load
- B) Edit, Transfer, Load
- C) Extract, Transfer, Log
- D) Execute, Transform, Load

Answer: A

4. What type of data does a data warehouse store?

- A) Operational data
- B) Historical data
- C) Real-time data
- D) All of the above

Answer: B

5. Which tool is commonly used for data analysis in data warehouses?

- A) OLTP
- B) OLAP
- C) XML
- D) JDBC

Answer: B

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6. Granularity in a data warehouse refers to:

- A) Size of the database
- B) Level of detail of data
- C) Number of dimensions
- D) Frequency of updates

Answer: B

7. The "snowflake schema" is an extension of which schema?

- A) Star schema
- B) Fact constellation
- C) Relational schema
- D) None of the above

Answer: A

8. Which is NOT a characteristic of a data warehouse?

- A) Integrated
- B) Time-variant
- C) Volatile
- D) Subject-oriented

Answer: C

9. In data warehousing, "drill down" means:

- A) Viewing data at a finer level of detail
- B) Aggregating data
- C) Removing redundant data
- D) Transforming data

Answer: A

10. A fact table contains:

- A) Dimensions
- B) Measures
- C) Both A and B
- D) None of the above

Answer: B

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11. Which type of schema has multiple fact tables?

- A) Star schema
- B) Snowflake schema
- C) Fact constellation
- D) None of the above

Answer: C

12. The process of summarizing data in a data warehouse is called:

- A) Drilling down
- B) Aggregation
- C) Transformation
- D) Extraction

Answer: B

13. Which phase of ETL involves cleaning the data?

- A) Extraction
- B) Transformation
- C) Loading
- D) Reporting

Answer: B

14. What does OLAP stand for?

- A) Online Analytical Processing
- B) Online Analysis Program
- C) Operational Layered Application Processing
- D) None of the above

Answer: A

15. What is a surrogate key in a data warehouse?

- A) A natural key
- B) A primary key
- C) A system-generated unique identifier
- D) None of the above

Answer: C

16. Which of the following is NOT an OLAP operation?

- A) Roll-up
- B) Drill-down
- C) Slice and dice
- D) Sort

Answer: D

17. Data mart is:

- A) A large warehouse for all company data
- B) A subset of a data warehouse
- C) Used for real-time data storage
- D) None of the above

Answer: B

18. The dimensional model is most commonly used for:

- A) Transactional systems
- B) Data warehousing
- C) Networking
- D) Real-time processing

Answer: B

19. Metadata in a data warehouse is:

- A) Data about data
- B) The raw data itself
- C) The summarized data
- D) None of the above

Answer: A

20. Which of the following ensures data integrity during ETL?

- A) Data profiling
- B) Transformation rules
- C) Primary keys
- D) All of the above

Answer: D

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21. What is meant by "slice and dice" in OLAP?

- A) Summarizing data
- B) Filtering and rearranging data
- C) Drilling down into data
- D) Removing redundant data

Answer: B

22. A slowly changing dimension (SCD) occurs when:

- A) Data remains static over time
- B) Dimension data changes slowly over time
- C) Fact data changes frequently
- D) None of the above

Answer: B

23. In star schema, dimensions are:

- A) Linked hierarchically
- B) Linked directly to the fact table
- C) Stored as raw data
- D) None of the above

Answer: B

24. Which type of OLAP uses pre-computed summaries of data?

- A) ROLAP
- B) MOLAP
- C) HOLAP
- D) None of the above

Answer: B

25. Factless fact tables are used to:

- A) Track events
- B) Store measures
- C) Contain only dimensions
- D) None of the above

Answer: A

26. What is a measure in a data warehouse?

- A) A metric of interest in the fact table
- B) A primary key in the dimension table
- C) A data transformation step
- D) None of the above

Answer: A

27. "Time-variant" in data warehousing means:

- A) Data is updated regularly
- B) Historical data is stored and maintained
- C) Data is integrated over multiple systems
- D) None of the above

Answer: B

28. Which of these is a valid example of a dimension?

- A) Product
- B) Sales amount
- C) Revenue
- D) Profit

Answer: A

29. A data warehouse architecture typically includes:

- A) Source systems, ETL processes, data warehouse, and BI tools
- B) Only operational systems
- C) Only BI tools
- D) None of the above

Answer: A

30. Which is a property of MOLAP systems?

- A) Uses relational databases
- B) Pre-aggregated multidimensional data cubes
- C) Does not support drill-down operations
- D) Slower performance compared to ROLAP

Answer: B

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31. Which is NOT an advantage of data warehousing?

- A) Improved decision-making
- B) Faster transaction processing
- C) Historical data analysis
- D) Data integration

Answer: B

32. Which operation in OLAP involves changing the level of data aggregation?

- A) Roll-up
- B) Drill-down
- C) Pivot
- D) Slice

Answer: A

33. Which is an example of a slowly changing dimension (SCD) Type 1?

- A) Overwriting old data with new data
- B) Maintaining historical data
- C) Creating a new record for every change
- D) Storing data in multiple tables

Answer: A

34. In a star schema, dimensions are connected to:

- A) Other dimensions
- B) Fact tables
- C) Metadata tables
- D) Operational data

Answer: B

35. Which layer of a data warehouse is used for reporting?

- A) Data staging layer
- B) Data access layer
- C) Metadata layer
- D) Operational data store

Answer: B

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36. What does a fact table typically store?

- A) Hierarchical relationships
- B) Quantitative data
- C) Metadata information
- D) Dimension keys only

Answer: B

37. A data warehouse is optimized for:

- A) Transactional data
- B) Analytical queries
- C) Real-time updates
- D) None of the above

Answer: B

38. Which of the following is NOT a data warehouse architecture type?

- A) Single-tier
- B) Two-tier
- C) Multi-tier
- D) Four-tier

Answer: D

39. Data redundancy in a data warehouse occurs due to:

- A) De-normalization
- B) Normalization
- C) ETL process
- D) Metadata storage

Answer: A

40. What does "schema-on-write" mean in a data warehouse?

- A) Defining schema before loading data
- B) Defining schema after data is queried
- C) Storing data without schema
- D) None of the above

Answer: A

41. Which of these is NOT an OLAP system type?

- A) ROLAP
- B) MOLAP
- C) SOLAP
- D) HOLAP

Answer: C

42. What is the purpose of a staging area in ETL?

- A) Transform data into the target format
- B) Store operational data
- C) Preprocess and cleanse data
- D) Generate reports

Answer: C

43. Data cubes in OLAP are used to:

- A) Store metadata
- B) Summarize data across dimensions
- C) Manage operational transactions
- D) None of the above

Answer: B

44. In data warehousing, "time dimension" refers to:

- A) Metadata storage
- B) Analyzing data across different time periods
- C) Real-time data analysis
- D) None of the above

Answer: B

45. Which key uniquely identifies rows in a dimension table?

- A) Foreign key
- B) Primary key
- C) Surrogate key
- D) Composite key

Answer: B

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46. "Drill-through" in OLAP refers to:

- A) Accessing transactional data from summarized data
- B) Aggregating data
- C) Creating new dimensions
- D) Removing duplicate records

Answer: A

47. Which architecture supports both ROLAP and MOLAP?

- A) Star schema
- B) Snowflake schema
- C) HOLAP
- D) None of the above

Answer: C

48. The main advantage of dimensional modeling is:

- A) Normalization
- B) Simplicity in query processing
- C) Reducing redundancy
- D) Real-time processing

Answer: B

49. Which of the following is NOT a characteristic of MOLAP?

- A) Uses multidimensional cubes
- B) Faster query performance
- C) Handles sparse data efficiently
- D) Relies on relational databases

Answer: D

50. The main purpose of metadata in a data warehouse is to:

- A) Store operational data
- B) Provide information about data
- C) Store summarized data
- D) None of the above

Answer: B

51. What does "pivot" in OLAP mean?

- A) Changing the view of data**
- B) Drilling down into data**
- C) Aggregating data**
- D) None of the above**

Answer: A

52. In a star schema, a fact table connects with:

- A) Dimension tables**
- B) Metadata tables**
- C) Summary tables**
- D) None of the above**

Answer: A

53. Which data warehouse architecture is easiest to scale?

- A) Single-tier**
- B) Two-tier**
- C) Multi-tier**
- D) None of the above**

Answer: C

54. An operational data store (ODS) is:

- A) A real-time data warehouse**
- B) Used for short-term decision-making**
- C) The same as a data mart**
- D) None of the above**

Answer: B

55. What is the purpose of normalization in a data warehouse?

- A) To improve query performance**
- B) To reduce redundancy in source data**
- C) To enhance aggregation**
- D) None of the above**

Answer: B

56. Which of these is an OLAP query type?

- A) CRUD operations
- B) Ad-hoc queries
- C) Aggregation
- D) Both B and C

Answer: D

57. The term "data granularity" refers to:

- A) Data accuracy
- B) Level of detail in data
- C) Storage capacity
- D) None of the above

Answer: B

58. Fact tables in a data warehouse are usually:

- A) Highly normalized
- B) Highly de-normalized
- C) Unstructured
- D) All of the above

Answer: B

59. Which is the best practice for designing a data warehouse?

- A) Use normalized schema
- B) Use denormalized schema
- C) Combine OLTP and OLAP systems
- D) None of the above

Answer: B

60. Which key is typically used in a fact table?

- A) Primary key
- B) Composite key
- C) Surrogate key
- D) None of the above

Answer: B

MCQS

Midterm preparation file

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Passing criteria of this course

Letter Grade	Grade Points	Equivalent Percentage
A+	4.00	90-100
A	4.00	85-89
A-	3.66 - 3.99	80-84
B+	3.33 - 3.65	75-79
B	3.00 - 3.32	71-74
B-	2.66 - 2.99	68-70
C	2.00 - 2.65	61-67
D	1.00 - 1.99	50-60
F	0.00	00-49

Passing Criteria for a Course:

The passing criteria is defined in such a way that it ensures the student shall be consistent in his studies throughout the semester.

Therefore, for passing a course/subject, student shall fulfill the following:

- Secure minimum **20%** score in Formative Assessments/Mid terms
- Secure minimum **20%** score in Final Term Examinations.
- Secure at least **40%** marks in aggregate while fulfilling the above requirements

Course Selection and Credit Hours

Q: How do I select courses when the course selection is open?

A: Follow these steps:

1. Check the credit hours allowed by your university (e.g., 21, 18, or 15 credit hours).
2. Each subject typically has 3 credit hours.
3. Divide the total allowed credit hours by 3 to determine how many subjects you can select.

Examples:

21 credit hours = 7 subjects
18 credit hours = 6 subjects

15 credit hours = 5 subjects

Remember, the university may allow different credit hours for each student, so check your specific allowance.

By following these steps, you can make informed decisions during course selection and manage your credit hours effectively.

ZB

Advice

Hmaesha relax ho ke parha kryn panic mat hua kryn apky parents ky bohat khawab hoty hn wo pura krny ki puri koshish kea kryn kbi b ksi pe depend na kea kryn apko bs ak insan success kr saqta ha or wo insan ap khud hn apky elawa koi nahe ap bs koshish kryn Allah pak pe strong yakeen rakha kryn or mehnat krke sb Allah ke hawaly kr dea kryn everything is possible be brave be strong stay blessed

Hum insan hamesha moat se darty or bhagty hn or moat se bachny ki koshish krty hn jabke moat ny ana hi ana ha hum

moat se nahe bach saqty humy jahanam se bachny ki koshish
krni chaheay hum jahanam se bach saqty hn
Insan ko 3 chezo se dar lagta ha Moat Risk/Dolat Ezat Shohrat
Fame

Moat

**Humy pta hona chaheay Moat tab ani ha jab Allah
chahy us se pehly puri dunya bi ak taraf ho ke apko
marna chahy to apka kuch nahe beggar saqti**

Risk/Dolat

**Risk Dolat sb Allah pak ke hath m ha puri dunya ak
taraf b ho jay na apsy apka risk cheen saqti ha na de
saqti ha Ye sirf ALLAH pak ky hath m ha**

Izat Zilat shohrat Fame

**Izat Zilat Sirf Allah de saqty hn puri dunya mel ke b
apko zra brabr b damage nahe kr saqti Agr ALLAH**

**Apko izat dena chahay to puri dunya ak taraf ho ke
b apko 1% b nuksan nahe pohancha saqti**

Phr Dar ks bat ka ????????????????

**Be brave be strong Just put your trust To ALLAH
Hamesha Confident or himat se raho kbi ksi k samne
mat jhuko puri dunya mel k b apka kuch b nahe
beggar saqti**

Tips

Payary Students kbi b Only files pe depend mat kea kryn just files read krke uspe total depend krna it's not Good Vu ke tamat Papers Conceptual hoty hn Agr apky concepts

Apny concepts clear kryn or handout ko read kryn

Agr Handouts lengthy

Lgy to Ap har topics Ki basic definitions dekh lea kryn or headings dekh leay kryn

Ap thori se mehnat Kryn bs ap kbi b fail nahe hongy jab himat haar jao to un maa baap k bare m socha kro Jo apki fees dete hn jinko ap pe bohat dawy hn Jo har jga kehty hn Hamara bacha university Mai parh rha ha un maa baap ki umeedy Kahi mat Tory apny kea naa Sahi apny maa baap k leay mehnat kro himat kro Everything is possible just do it my Family

Apko Kuch b samj naa ay yaa ap Kuch b pochna चाहy to ap mujsy contact kr saqty hn

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Mri tamaat files Assignment GDB Solutions files Quiz etc. A to Z Sb free of cost ha Vu k bohat se log mere name pe passy lete hn Mera unsy koi lena dena nahe so be aware from them ZB Jo b krta ha Allah ki mehlok ki khudtam k leay krta ha I don't need anything from anyone in return because I am doing this business with My Allah

ZB
MY REQUEST FROM ALL OF YOU

ZB Request from All of you My family

Mjy ap sbki help or support ki zarort ha or wo ye k hum sb Mel kr Allah ka Quran ki Urdu translation logo tak pohanchy or Quran ko samjna asaaan kryn dosro k leay sb tak Allah ka Quran pohanchy it's my campaign hum log 70+ age k ho jaty hn phr b hmy namaz tak ki translation nahe pta hoti k hum Allah pak se

Kya Dua kr rhy hn so hum youngster's ko Mel kr puri takat or energy ke sath Allah ka Quran spread krna ha

Hum puri Koshish kr rhy hn k hum sb Mel k Quran ki translation or most important topics ko maximum share kry with translation hmara maksad Quran ko spread krna ha be a part of us

m apni pocket se ye sb kr rha ho Allah ka Quran spread krny k leay apne or mene sbny Marna ha ak din to q na Allah k leay Kuch kryn apni energies apni power ko bajay negative use krny k Allah k leay invest kryn khud ko be a part of us

Agr m 23 ki age m apny sare sources use kr rha ho pocket se heavy amounts give away kr rha ho Quran spread krny k leay to ap just share to kr saqty hn itna e krdyn

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Agr ap mujsy contact na kr sky too b ap jo b hn jaha bi hn waha Allah ka quran pheelay zada se zada with urdu translation jitni himat ha utna share kryn

..... **MY Family**

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..... **AL-Quran**

"And indeed, with hardship comes ease." (Quran 94:5)
Remember, I'm here to support you! Stay brave and strong!"

May Allah bless you